AI-Driven Inventory Optimization

Capstone Project: Stock CoPilot

Store managers are often forced to guess what to reorder, when to reorder, and how much to buy. That guesswork leads to empty shelves, excess stock, and too much time spent reacting instead of planning.

MONTE CARLO SIMULATION | PORTFOLIO OPTIMIZATION

CONVERSATIONAL AI FOR SUPPLY CHAIN ORDERING

Why it matters

- Keep shelves stocked without overbuying
- Cut waste and markdowns
- Save time on ordering decisions

Team

- Kaiwalya Mishra
- Aditya Chache
- Dev Patel



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The Business Challenge

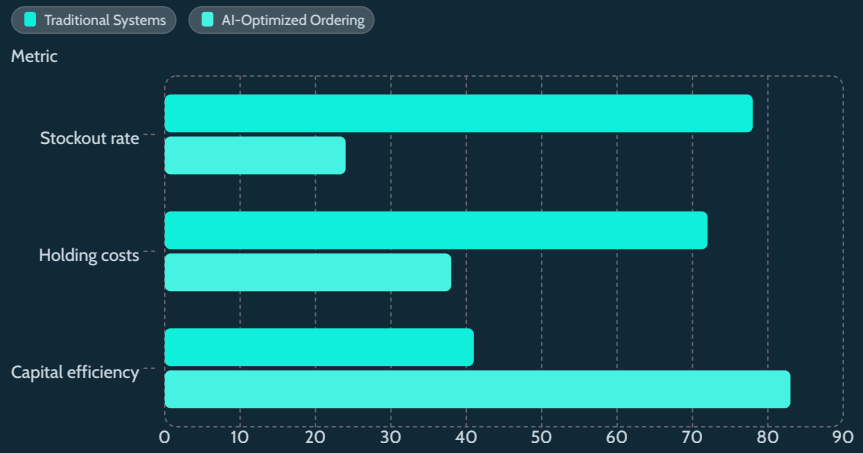
Retail teams must keep shelves full without tying up too much cash in inventory. That balance gets harder as demand shifts, costs rise, and every SKU competes for limited attention.

Service Level
Prevent stockouts and protect customer satisfaction.

Cost Efficiency
Limit holding, ordering, and logistics costs.

Capital Allocation
Put working capital behind the highest-priority SKUs.

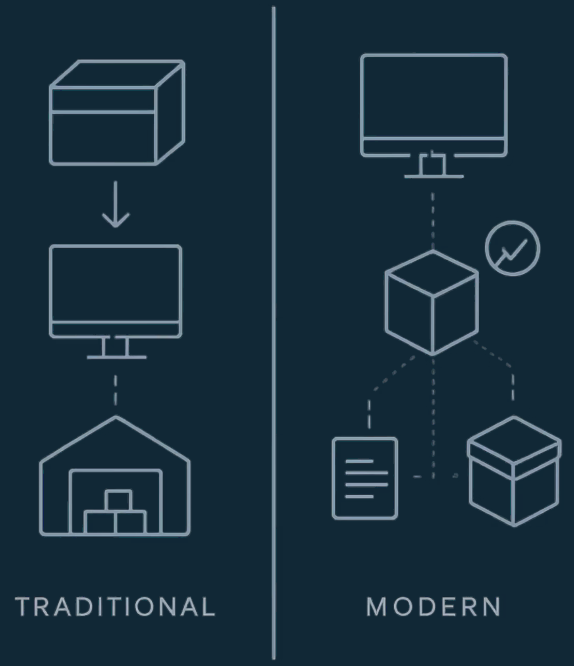
The Gap
- Traditional systems often rely on fixed rules that miss demand variability.
- They struggle to weigh stockouts, excess inventory, and capital trade-offs at the same time.
- They provide recommendations, but not the flexibility to adapt as conditions change.



- **Lower stockouts:** AI-optimized ordering performs better on availability.
- **Lower carrying cost:** Less excess inventory reduces holding costs.
- **Better capital use:** Inventory dollars are allocated more efficiently.



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Traditional Systems Fall Short

Retail demand is too variable for static, rule-based tools to manage well. Teams need a faster, more flexible way to balance service levels, risk, and capital.

What's Missing Today

- Static reorder rules miss demand variability
- No simulation to test operational risk
- No portfolio-level capital optimization
- No clear, manager-friendly guidance

The Shift

- From fixed rules to confidence-based forecasts
- From guesswork to Monte Carlo risk visibility
- From SKU-by-SKU decisions to portfolio optimization

What Our Solution Delivers

- Forecasts with P10, P50, and P90 bands
- Monte Carlo simulation for risk visibility
- Optimization across hundreds of SKUs
- Conversational AI for rapid insights

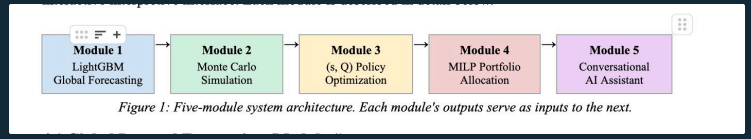


Figure 1: Five-module system architecture. Each module's outputs serve as inputs to the next.

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Dataset & Forecasting Foundation

Walmart M5 matters because it mirrors the messy reality of retail demand: thousands of SKUs, changing prices, holidays, and uneven buying patterns. It gives us a realistic testbed for building forecasts that can handle noise, seasonality, and store-level variation. Our model uses five years of sales history across three states to learn these patterns at scale.

200

Selected SKUs

Representative products sampled across CA_1, TX_1, and WI_1.

600

Time Series

SKU-store combinations used to learn local and product-level demand.

5

Years of Data

Historical sales from 2011–2016, including holdout validation.

Best Model: LightGBM

- Captures nonlinear demand patterns and feature interactions
- Handles noisy retail data and intermittent sales well
- Uses price, seasonality, and lagged sales signals effectively
- Generalizes strongly on held-out 2016 data

Forecast Output

- **P10**: conservative downside scenario for risk planning
- **P50**: median forecast for core inventory decisions
- **P90**: upside scenario to reduce stockout risk

These bands create a practical planning range instead of a single point estimate.

Dataset Scope

- 5 years of daily sales history
- 3 states: CA, TX, and WI
- Thousands of products and store combinations
- Holidays, pricing, and demand volatility included

What This Enables

- Realistic retail forecasting at scale
- Better signal for uncertain demand
- More reliable decisions on inventory and service levels

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OPTIMIZATION LAYER

Single SKU Optimization

One SKU may seem small, but across hundreds of stores and products, the right replenishment policy adds up fast. This model finds the lowest-cost order strategy while still protecting shelf availability.

Objective

- Minimize total inventory cost: holding cost + ordering cost + stockout penalty
- Balance profitability with product availability
- Avoid overstocking and unnecessary emergency replenishment

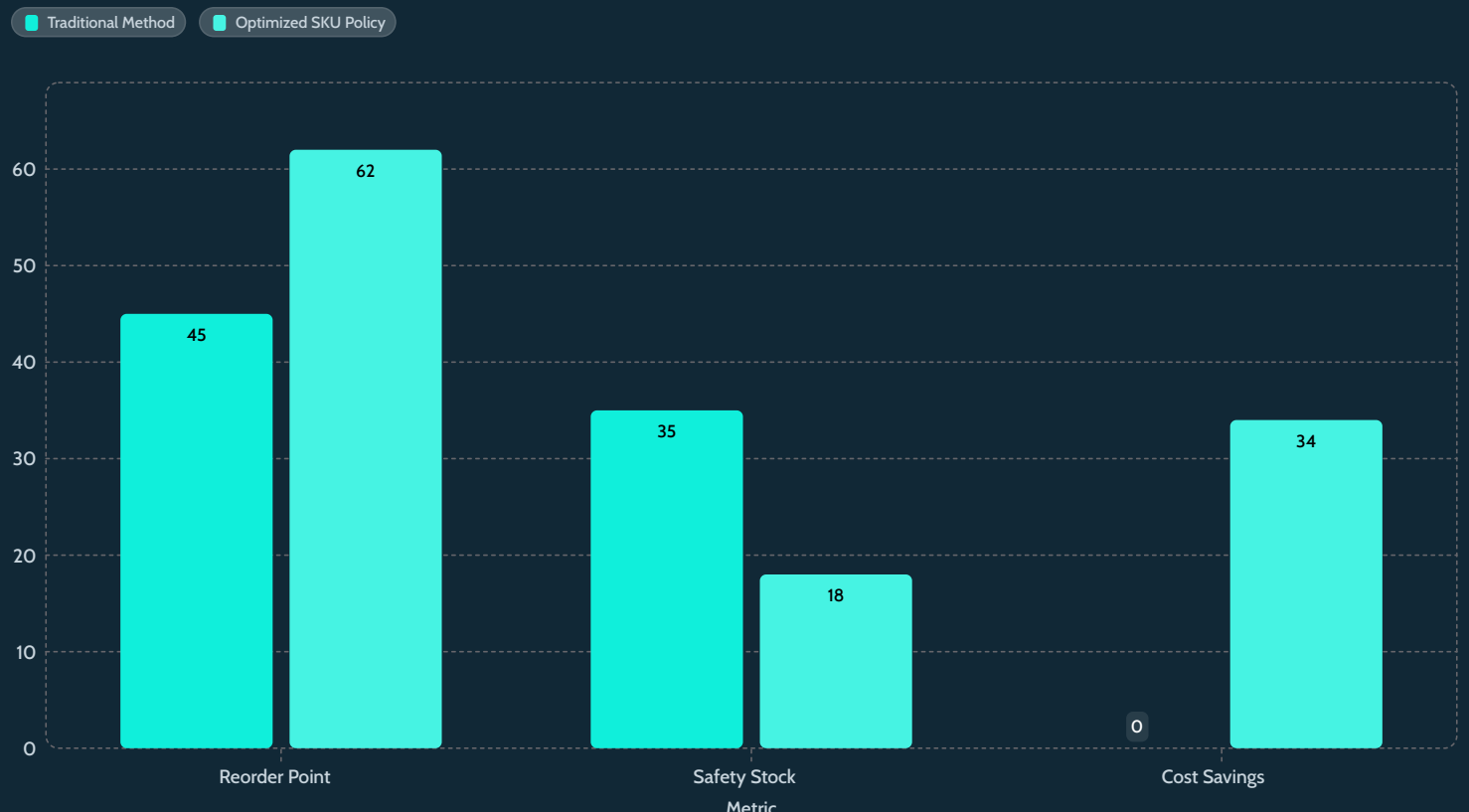
Decision Variables

- **Reorder Point (s)**: the inventory level that triggers a new order
- **Order Quantity (Q)**: how many units to reorder each time
- Higher **s** protects against demand spikes; **Q** controls how much inventory is added

Constraint

- Keep the fill rate at or above the target service level, such as 95%
- Protect customer availability while avoiding excess cost

Optimization Results



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Portfolio-Level Optimization: Multi-SKU Analysis

Portfolio optimization looks at inventory as one connected system, not hundreds of separate decisions. By evaluating SKUs together, the model finds stocking policies that improve overall performance, reduce waste, and allocate capital where it creates the most value.

- Analyzes hundreds of SKUs simultaneously to find globally optimal inventory policies.
- Reallocates working capital toward high-value and high-volatility items with the strongest return.
- Balances service levels across categories so critical items stay available without overstocking others.
- Creates a harmonized inventory plan that improves efficiency across the full portfolio.

The Process

For each SKU, we generate feasible (s, Q) policies using forecasting and Monte Carlo simulation. Each policy has four key metrics: Expected Operating Cost, Working Capital Required, Procurement Spend, Service Level.

- We formulate a constrained optimization problem: select exactly one policy per SKU to minimize total operating cost while keeping total working capital within budget (e.g., $3,000 or $5,000)

What This Reveals

Capital Investment
Which SKUs deserve more capital investment

Operating Cost
Where operating cost is concentrated across the portfolio

Service Level Risk
Where service level risk is concentrated

Budget Utilization
Whether the budget is underutilized or tight

This transforms inventory from a collection of independent decisions into a strategic capital allocation problem, helping managers invest where it matters most.

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AI Inventory Strategy Assistant

An **LLM-powered conversational interface** turns complex optimization results into clear guidance managers can use right away. It helps store teams make faster inventory decisions without needing data-science expertise.

Conversational Interface

Ask inventory questions in plain English and get direct, structured answers.

Smart Reasoning

Understands shocks, follows up naturally, and keeps context across questions.

Actionable Insights

Explains tradeoffs and recommends reorder changes managers can act on.

AI Assistant Capability Snapshot

- Answers manager questions with plain-language recommendations
- Summarizes optimization outputs into simple next steps
- Highlights capital-intensive SKUs and service-level tradeoffs
- Supports follow-up questions with conversation context

How It Works

- Manager asks a question about inventory or replenishment
- LLM interprets the request and routes it through the optimization logic
- The assistant returns concise guidance with recommended actions

Built with: Claude Sonnet 4.6 LLM and a rule-based routing engine.

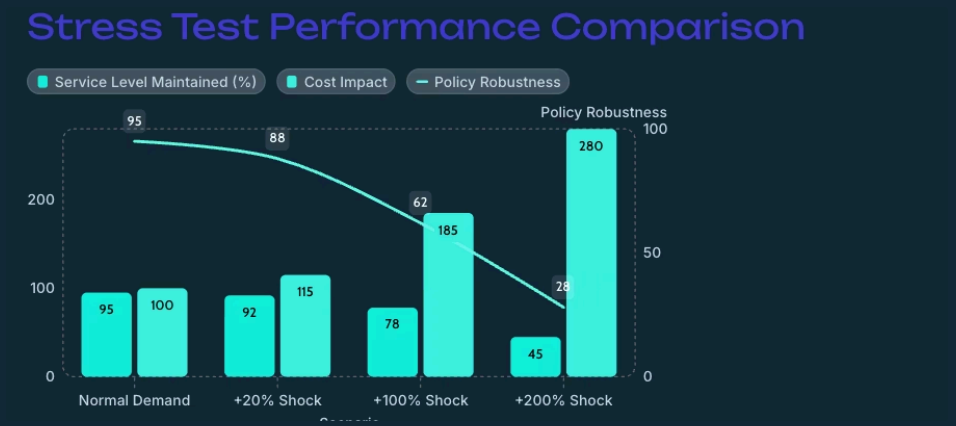


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Scenario Stress Testing: 75% Demand Shock

Shock analysis evaluates how an inventory policy performs when demand changes abruptly and substantially. It matters because it reveals whether a system can preserve service levels, control costs, and remain operationally resilient under conditions that are more volatile than normal planning assumptions.

Comparison Table: Baseline, Shocked, and Shock-Optimized Policies



Key Findings

Fill Rate

Baseline holds **99.96%**; shock-optimized reaches **100%**.

Cost

Shock cost rises from **$60** to **$392** optimization lowers it to **$325**

Risk

Low risk overall—resilient under extreme demand variability.

Conclusion

Overall, the 75% demand shock results show a robust inventory system that preserves near-perfect service while limiting the downside of volatility. The comparison suggests that optimization improves both resilience and cost control, making the policy better suited for uncertain demand environments.

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Managerial Impact & Real-World Value

Built for a **Walmart store manager**, this system turns model outputs into clear action. It helps leaders make faster decisions on inventory, spending, and service levels with more confidence.

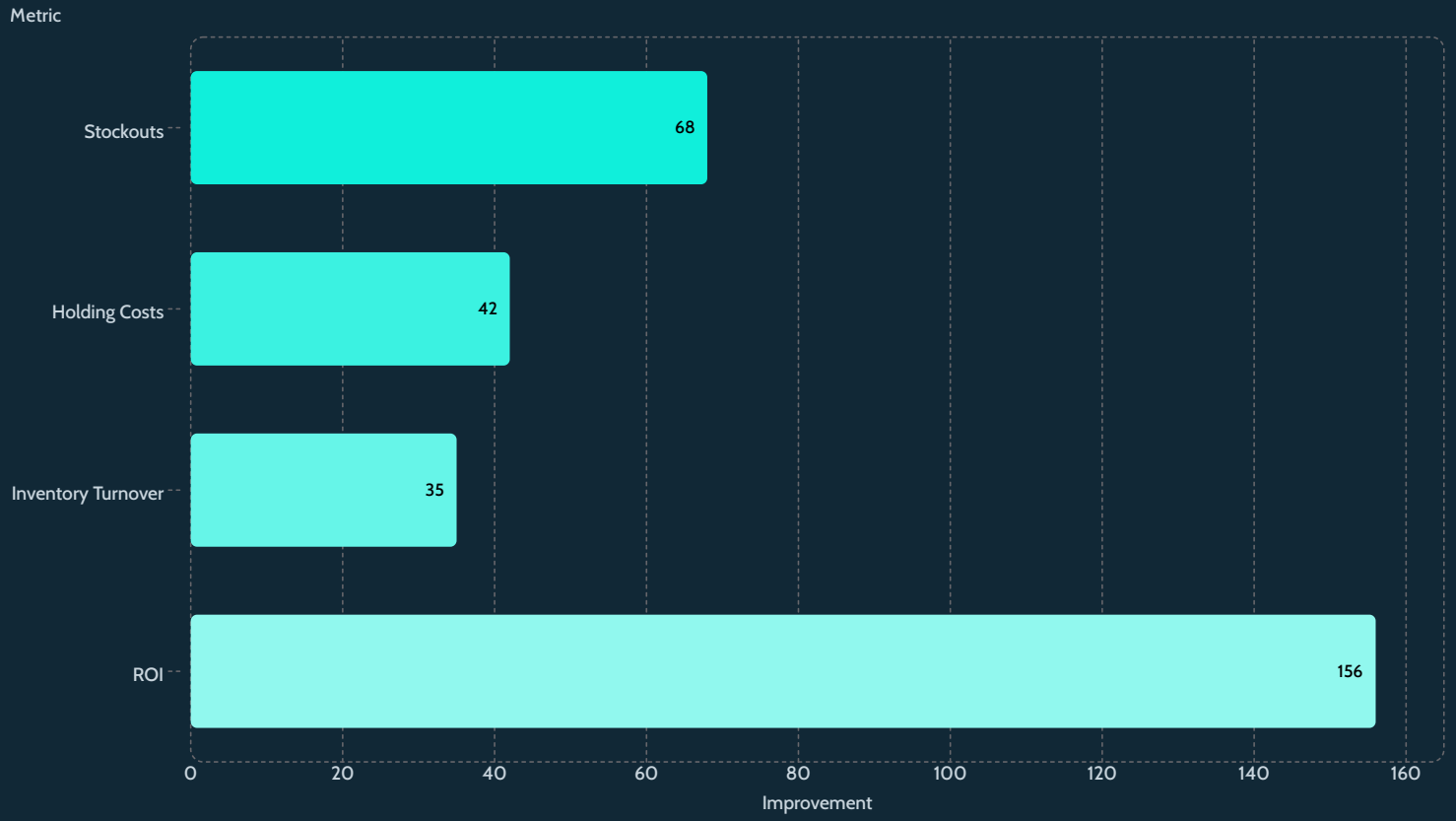
Risk-Aware Decisions
Monte Carlo simulation makes uncertainty visible so managers can act on ranges, not guesses.

Capital Efficiency
Optimization shifts working capital to higher-priority SKUs and reduces waste.

Scenario Planning
Stress tests show how costs and service levels change under demand shocks.

Executive Support
Conversational AI turns complex outputs into boardroom-ready recommendations.

Managerial Impact Metrics



- Fewer stockouts improve shelf availability and customer satisfaction.
- Lower holding costs free up capital for more important needs.
- Higher inventory turnover and ROI strengthen overall store performance.

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Team & Contributions

Capstone Project: AI-Driven Inventory Optimization

Kaiwalya Mishra

- Exploratory data analysis on Walmart M5 dataset
- Forecast model development and optimization
- Agent development for demand prediction
- Model validation and performance testing

Aditya Chache

- Simulation engine development
- Portfolio optimization algorithm design
- Streamlit backend implementation
- System integration and performance optimization

Dev Patel

- Conversational AI assistant implementation
- UI/UX design and development
- Backend integration for chat interface
- User experience optimization

Project Impact

- Successfully integrated advanced data science, optimization, and AI to create a comprehensive inventory management solution.
- Delivered a user-friendly system that simplifies complex inventory decisions for retail managers.
- Demonstrated significant potential for reducing stockouts, lowering holding costs, and improving capital efficiency.
- Provided robust stress-testing capabilities to ensure policy resilience under various demand scenarios.

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